



DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

AWS STUDENT BUILDER GROUP

The RAG Builder Session

3-3.5 HOURS · BEGINNER-FRIENDLY · 22 OCT 2026

Welcome to The RAG Builder Session

AWS SETUP → S3 + RAG → KNOWLEDGE BASE → BUILD & TEST

PREREQUISITES

What You Need to Bring

- A **laptop with a stable internet connection**.
- No prior experience with **AWS, Artificial Intelligence, Generative AI, or RAG** is required.
- The workshop is **beginner-friendly**, and all required concepts will be introduced during the session.

WORKSHOP BRIEF

Build Practical Skills in Generative AI and AWS

BEGINNER-FRIENDLY // AWS // GENERATIVE AI // HANDS-ON

The workshop introduces participants to AWS account setup, Amazon Bedrock, Knowledge Bases, and Retrieval-Augmented Generation (RAG) through a hands-on learning experience. Participants will learn how to set up their AWS environment, connect documents with AI, retrieve relevant information, and build their own AI-powered Knowledge Base. It focuses on developing practical skills in generative AI and AWS services.

SESSION 1

AWS Fundamentals & Account Setup

- Introduction to AWS, its core services, and the AWS Cloud ecosystem.
- Hands-on AWS account setup and basic console navigation.

CHECKPOINT: Participants successfully set up their AWS account and become familiar with the AWS Management Console.



SESSION 2

Amazon S3, AI & RAG Fundamentals

- Introduction to Amazon S3 and hands-on bucket creation.
- Overview of Artificial Intelligence, Generative AI, and Retrieval-Augmented Generation (RAG).

CHECKPOINT: Participants create an Amazon S3 bucket, upload a sample document, and understand the basic concepts of AI and RAG.



SESSION 3

Building a RAG Knowledge Base

- Hands-on session on Amazon Bedrock and Knowledge Bases.
- Set up a RAG-based Knowledge Base using documents and learn how to retrieve relevant information.

CHECKPOINT: Participants successfully create and test their own RAG-based Knowledge Base using Amazon Bedrock.

KEY TAKEAWAYS

- Understanding of AWS fundamentals and the AWS Cloud ecosystem.
- Ability to set up and navigate an AWS account and Management Console.
- Hands-on experience with Amazon S3, including creating buckets and uploading documents.
- Basic understanding of Artificial Intelligence, Generative AI, and RAG.
- Knowledge of how Retrieval-Augmented Generation (RAG) works.
- Hands-on experience with Amazon Bedrock and Knowledge Bases.
- Ability to build, synchronize, and test a RAG-based Knowledge Base.
- Practical foundation for participating in Beyond Search: The Intelligence Challenge.

BUILD, SYNCHRONIZE & TEST YOUR OWN RAG-BASED KNOWLEDGE BASE

WORKSHOP GUIDE

THE RAG BUILDER SESSION // INFORMATION SCIENCE AND ENGINEERING // 22 OCT 2026

SESSION	TIME	TOPIC	OUTCOME
Session 1	—	AWS Fundamentals & Account Setup	Set up an AWS account and navigate the Management Console.
Session 2	—	Amazon S3, AI & RAG Fundamentals	Create an S3 bucket, upload a sample document, and understand AI and RAG basics.
Session 3	—	Building a RAG Knowledge Base	Create and test a RAG-based Knowledge Base using Amazon Bedrock.

The RAG Builder Session

A hands-on workshop covering AWS environment setup, Amazon Bedrock, Knowledge Bases, and Retrieval-Augmented Generation. Participants connect documents with AI, retrieve relevant information, and build their own AI-powered Knowledge Base.

QUICK OVERVIEW — THE FLOW

3-3.5 HOURS

3 HANDS-ON SESSIONS

BUILD & TEST A RAG-BASED KNOWLEDGE BASE

KEY TAKEAWAYS

Understanding of AWS fundamentals and the AWS Cloud ecosystem.

Ability to set up and navigate an AWS account and Management Console.

Hands-on experience with Amazon S3, including creating buckets and uploading documents.

Basic understanding of Artificial Intelligence, Generative AI, and RAG.

Knowledge of how Retrieval-Augmented Generation (RAG) works.

Hands-on experience with Amazon Bedrock and Knowledge Bases.

Ability to build, synchronize, and test a RAG-based Knowledge Base.

Practical foundation for participating in Beyond Search: The Intelligence Challenge.

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